Genealogical Priors in Self-Supervised Learning: Improving Speech Technology for Low-Resource Languages

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Abstract

Over 7,000 languages are spoken worldwide, yet speech technology focuses on high-resource ones, leaving many underrepresented. Current strategies prioritize scaling data and compute, often overlooking linguistic structure. Motivated by evidence that related languages share phonetic features, we examine whether Self-Supervised Learning (SSL) models can leverage genealogical relationships to improve transfer to low-resource languages. Using WavLM-Large, we compare random multilingual training with language-family-aware subsets. Family-aware training outperforms random selection on downstream tasks despite similar pretraining convergence. However, both remain below the original WavLM-Large baseline, indicating that linguistic structure enhances robustness under data constraints but complements rather than replaces large-scale pretraining. Incorporating linguistic organization into SSL offers a principled path toward more inclusive speech technology.

Index Terms: language family, low-resource languages, SSL

1. Introduction

Languages are not static systems: they evolve, diverge, and influence one another across generations and geography. Historical linguistics has long established that languages change through regular sound and branching descent from common ancestors [1]. Related languages within each family often retain systematic correspondences in phonology, morphology, and broader typological features due to shared inheritance [2]. Using data from the World Atlas of Language Structures (WALS) [3], a study demonstrated that certain features of languages, such as lexicon and word order, remain highly stable within families, while others fluctuate due to geographic contact rather than heritage [4].

Although the world exhibits rich linguistic diversity, the vast majority of the approximately 7,000 languages spoken today remain severely underrepresented in speech technology [5]. Most are low-resource languages, which by definition [6] are "less studied, resource scarce, less computerized, less privileged, less commonly taught or low density", placing them at risk of digital marginalization. In speech technology, the study of cross-linguistic structure is fundamentally constrained by the scarcity and cost of annotated speech data. High quality transcriptions require expert knowledge, especially in low-resource languages, and are time-consuming. This imbalance has historically limited supervised approaches, as demonstrated by Zhong et. al [7], Ranathunga et. al [8], and David et. al [9] in their different studies related to Low-Resource languages representation in speech technology. The emergence of Self-Supervised Learning (SSL) has significantly altered this landscape.

The introduction of Wav2Vec 2.0 [10] marked a turning point in speech representation learning, demonstrating that contrastive predictive objectives on raw speech could achieve state-of-the-art performance across downstream tasks. This paradigm spurred subsequent work scaling wav2vec architectures to larger models and massively multilingual settings [11, 12, 13, 14], alongside the emergence of alternative SSL architectures such as HuBERT [15] and Data2Vec [16]. These models, however, share a common characteristic in how training data is handled: they rely on continuous audio streams without an explicit strategy for sampling or organizing data based on linguistic structure. Consequently, multilingual diversity is treated primarily as a scaling variable rather than as a structured source of inductive bias, with the data pipeline focusing on format and scale over linguistic considerations.

Rather than exposing a SSL model to an arbitrary multilingual mixture, we constructed training sets composed of multiple families, selecting two to three languages per family when available. Our central hypothesis is that this structured exposure encourages the model to learn family-level phonetic regularities that generalize beyond the languages seen during training. To evaluate this, we pretrained WavLM-Large model [17] on 35 languages spanning diverse families and compare family-aware sampling against random multilingual selection. The resulting models, submitted as embedding extractors under the Interspeech Unsupervised People's Speech in the Wild challenge, are evaluated on unseen languages across downstream tasks including Automatic Speech Recognition (ASR), language identification (LID), and speaker clustering. Our findings show that linguistic structure provides a meaningful inductive bias, with family-aware pretraining outperforming embeddings learned through unstructured multilingual scaling.

2. Methodology

2.1. Data and Preprocessing

The Unsupervised People's Speech in the Wild challenge, featured at Interspeech 2026, provides the dataset for our experiments. The Unsupervised People's Speech dataset [18] contains over 1M hours of audio. To ensure data quality, the authors performed two preprocessing analyses: Voice Activity Detection (VAD) using Silero's VAD pipeline, which identified 820k+ hours of actual speech, and Language Identification using Nvidia's TensorRT-LLM implementation of Whisper Large v3, which detected 89 languages across the corpus. More details about the statistics of the data can be found in the dataset card from Hugging Face [**?**].

Building on these annotations, we applied additional preprocessing criteria to prepare the data for our self-supervised learning pipeline:

- **Segment duration filtering:** We used at least 16 segments of minimum 10 seconds per audio file to enable meaningful feature extraction. Analysis of the VAD results showed that 97.29% of audio files did not meet this criterion.
- **Language availability filtering:** We excluded audio files where both the VAD and language detection were empty. This applied to $192,872$ audio files.
- **Final usable data:** After applying both filters, only $262,275$ audios met our criteria, while $1,981,063$ audios were discarded.

The resulting subset of audio still amounted to $+281$K hours of speech, which was sufficient for both our available computational resources and the scope of the study. Prior to final preprocessing, we conducted exploratory analyses to examine whether short or weakly segmented audio influenced the geometry of the learned representations. Using audio samples from 25 different languages (with one to two representative languages per language family), we compared embeddings extracted from the same audio files under two conditions: (i) without applying VAD-based segmentation, and (ii) after applying our VAD and segment-duration constraints. Embeddings were obtained using pretrained Wav2Vec-base and HuBERT-base models.

Using these embeddings, we quantitatively assessed how the different preprocessing conditions affected representational structure. Clustering consistency in the embedding space was evaluated through a nearest-centroid classification procedure. For each language, we computed the mean embedding (centroid) across its samples. Each embedding was then assigned to the language whose centroid was closest in Euclidean distance, and accuracy was computed as the proportion of correct assignments. The same procedure was repeated at the language-family level. Results are shown in Figure 1.

Table 1: *Model performance with and without VAD preprocessing*

Model	Condition	Language Acc	Family Acc
HuBERT	no VAD	0.56	0.43
HuBERT	VAD	0.65	0.50
Wav2Vec	no VAD	0.43	0.30
Wav2Vec	VAD	0.50	0.37

Across both pre-trained models, VAD-based filtering consistently improved centroid classification at both the language and family levels. For HuBERT, language-level accuracy increased from 55.9% to 65.2%, while family level accuracy increased from 42.5% to 50.0%. Similar behavior was found at language and family-level accuracies. These results indicate that VAD filtering leads to more compact and better-separated clusters in embedding space.

2.2. Building language families

The core of our approach relies on accurately mapping the 89 languages found in the dataset to their respective families. This mapping shown in Appendix A, Table 3, was constructed using established linguistic resources such as Glottolog [19] and Ethnologue [20]. The distribution of audio segments in the full dataset is highly imbalanced (Figure 2), with Indo-European accounting for approximately 88% of all available speech segments. However, the selection of the training subset was performed at the language level, not at the audio level. Our objective was to ensure diversity rather than to replicate the raw audio imbalance of the corpus. Therefore, larger families were represented by a greater number of languages, while smaller families were included in their entirety when only one language was available. When possible, we ensured a minimum of two languages per family to allow for within-family variability during training. The final training dataset consisted of 32 languages across multiple families, comprising ≈ 56 hours of audio. The distribution of this dataset This dataset will be referred to as "language-aware" from now on.

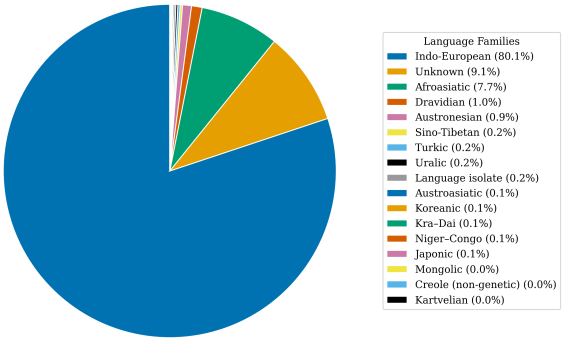

Figure 1: *Distribution of selected data after VAD and preprocessing steps*

For the "no-language-aware" evaluation, we constructed a training set by randomly sampling audio files from the UPS dataset without any language or family-based filtering. To ensure comparability, this subset was matched in total duration to the language-aware configuration.

2.3. Models

In recent years, SLL speech models have substantially advanced the extraction of the contextualized representations directly from raw speech signals, enabling models to learn rich acoustic structure without reliance on large-scale labeled data. In this work, we use WavLM-Large, a 316M-parameter transformer-based self-supervised model with relative positional attention, trained through masked speech prediction and denoising objectives [17]. Rather than using it off-the-shelf, we perform continued pre-training on UPS data with a contrastive objective and Gumbel-Softmax vector quantization. The convolutional feature encoder is frozen to preserve well-learned low-level acoustic features, while the transformer encoder and projection heads are updated. WavLM's strong performance across diverse downstream tasks on the Speech processing Universal PERformance Benchmark (SUPERB) [21] indicates that its representations capture transferable acoustic information, making it an appropriate backbone for the UPS challenge.

2.4. Experiments

Two experiments were conducted under identical training parameters, using the no-language-aware and language-aware subsets. For both setups, fixed-duration chunks were randomly extracted from each audio file, using approximately 60 hours of audio in total. Chunks of a fixed duration were extracted inde-

pendently from each audio file. The number of chunks extracted per file scaled proportionally with its duration, such that longer recordings contributed more chunks in absolute terms. However, a minimum of 8 chunks per file was enforced regardless of duration, meaning that shorter files were sampled more densely relative to their actual length. Consequently, while sampling was uniform within each file, it was not strictly uniform across the corpus in terms of audio-duration contribution, as shorter recordings were relatively overrepresented on a per-second basis.

Audio was resampled to 16kHz, converted to mono, normalized using the feature extractor of the WavLM-Large model, and trimmed or zero-padded to a fixed length. During training, on-the-fly waveform-level augmentations were applied exclusively to the training subset, consisting of stochastic speed perturbation and additive noise. A detailed description of these augmentations is provided in Section 2.5. These techniques encourage robustness to natural variations in speaking rate and background acoustic conditions [22], a design principle that mirrors the pretraining objective of WavLM itself.

For the validation set, we used a dedicated held-out dataset constructed under the same preprocessing pipeline, with the exception that no data augmentation was applied during training. This split contained 27 files of 10 different languages not seen during pretraining, selecting, when possible, one unseen language per family. Distribution of this dataset is presented in Appendix A, Table 4.

2.5. Experimental Setup

The WavLM Large model was fine-tuned using a self-supervised contrastive pretraining objective on a multilingual speech corpus. The training set yielded $\approx 22,000$ chunks, while the validation set consisted of $\approx 7,000$ chunks. Each chunk was fixed at 10 seconds, randomly extracted from its source file by sampling a start position uniformly at random within the valid range, with a minimum of 8 chunks extracted per file. Training was conducted for 8 epochs with a per-device batch size of 2 and gradient accumulation over 8 steps, resulting in an effective batch size of 16. The Adam W optimizer was used with $\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 1e-6$, and weight decay of 0.01. A cosine learning rate schedule with 500 warmup steps was applied, with an initial learning rate of 5e-5. Gradient norms were clipped at 10.0. Mixed precision training in bfloat16 was used throughout. The CNN feature encoder of WavLM was kept frozen during training to preserve low-level acoustic representations, while the transformer layers and pretraining heads were updated. A time masking probability of 0.65 was applied following the default WavLM pretraining configuration. The Gumbel softmax temperature was decayed exponentially from 2.0 to 0.1 with a decay factor of 0.999998 per step. Two data augmentation strategies were applied exclusively to training chunks: speed perturbation with a factor drawn uniformly from $[0.9, 1.1]$ (applied with 50% probability), and additive white Gaussian noise at a signal-to-noise ratio drawn uniformly from $[15, 30]$ dB (applied with 30% probability). No augmentation was applied to the validation set. All experiments were run on a CUDA-enabled GPU using Google Colab.

3. Results

The language-aware and no-language-aware configuration converged successfully and achieved final validation losses of 3.86 and 3.93 respectively, corresponding to total improvements of 10.3% and 9.6% from their initial values for the 60h training.

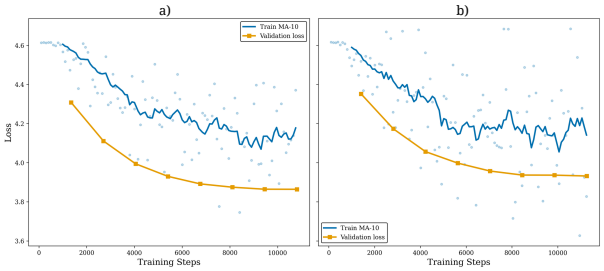

Figure 2: *Total loss of a) Training on language-aware data; b) Training on no-language-aware data*

Table 2: *WavLM model performance on UPS challenge downstream tasks for language-aware and no-language-aware subsets trained on 60 hours of data*

Condition	Score	F1	CER	ARI
language-aware	0.50	0.50	0.75	0.48
no-language-aware	0.04	0.037	0.90	0.39

Despite comparable pretraining convergence, the two configurations yielded substantially different results on the LID, ASR, and Speaker Clustering tasks, as evaluated in the UPS Challenge in the Wild and shown in Table 2. The language-aware configuration achieved a general score of 0.50, while the no-language-aware configuration obtained 0.04. This disparity, despite similar SSL dynamics, suggests that the composition and selection strategy of the pretraining data have a stronger influence on downstream performance than the convergence profile alone. Under severe data constraints (60h), continued pretraining appears to act less as representation expansion and more as perturbation: structure-aware sampling mitigates degradation, whereas random sampling amplifies it, leading to near-collapse in downstream performance.

However, the official WavLM-large baseline substantially outperforms both continued-pretraining models. Given that the model was originally pretrained on approximately 94k hours of curated English speech, our 60h continued-pretraining regime is likely insufficient to meaningfully restructure a 300M-parameter model. While the additional training was enough to modify the loss trajectory, it may have partially perturbed well-optimized English representations without providing sufficient multilingual evidence to build stable cross-linguistic abstractions.

4. Discussion

Our results suggest that, under constrained data conditions, intelligently curated data can outperform indiscriminately scaled data. Similar findings have been reported in underwater acoustics [23], speech enhancement [24], and other domains [25], where curated subsets surpassed larger randomly sampled datasets of identical or greater size. This pattern aligns with how SSL speech models learn, as representation quality depends critically on the composition of negative pairs encountered during contrastive training [26, 27]. Random multilingual

sampling likely produces either trivially easy or false-negative pairs [28], limiting discriminative pressure. This mechanism offers a plausible explanation for the large performance gap observed in the LID task.

However, improvements were not limited to LID. The language-aware configuration achieved substantially lower CER in ASR and higher ARI in speaker clustering. The CER gains suggest that cleaner input distributions facilitate more stable acoustic representations, consistent with prior findings that curated or noise-controlled data improves SSL robustness [24]. The ARI improvement may reflect two complementary effects: (1) reduced acoustic noise and (2) increased structured diversity across language families, which encourages the model to learn more separable speaker representations. Since SSL objectives depend on meaningful negative sampling [26, 27], structured multilingual diversity likely enhances clustering-friendly embeddings compared to random heterogeneous sampling.

A complementary explanation for this gap is catastrophic forgetting (CF) in the "no-language-aware" subset [29]. CF refers to the tendency of neural networks to degrade previously learned representations when exposed to new or mismatched data distributions [30]. This is particularly relevant here: WavLM was originally pretrained on English speech, while our continued pretraining uses the UPS dataset, a multilingual corpus spanning 89 languages with variable audio quality and no VAD preprocessing for this specific subset. The resulting data is both acoustically heterogeneous and potentially contaminated by non-speech segments. Wei-Ning et al [31] showed that domain mismatch between pretraining, fine-tuning, and evaluation significantly degrades SSL speech performance, supporting the importance of data curation and controlled input distributions during multilingual pretraining.

The superior performance of the language-aware configuration can therefore be attributed to its principled structure. By selecting languages across different families, the training corpus imposes a hierarchical organization over the acoustic space. This structured coverage resembles curriculum learning (CL), where organizing training data according to meaningful relationships yields more stable and transferable representations [32, 33]. In SSL speech models, curriculum-based strategies have improved ASR [34], speaker verification [35], and end-to-end speech systems [36, 37, 38]. Our results suggest that language-family organization could function as a form of implicit curriculum over linguistic structure.

While these findings are promising, they must be interpreted within the constraints of our experimental design. We acknowledge that our experiments were conducted with a single and small subset of audio data. Therefore, a more comprehensive study involving varying model sizes and data scales is necessary to fully validate the generalization of this data-centric approach.

Finally, while genealogical structure provides a powerful organizing principle [2], language evolution is not purely tree-like. Thomason (2001) [39] established that geographic proximity and language contact drive structural convergence across unrelated languages. Törö et al. (2025) [40] demonstrate that embedding distances correlate with both genetic and geographical distance, with minority languages often clustering with geographically dominant ones rather than genetic relatives. Together, these findings suggest that SSL models encode both deep genealogical relationships and more recent contact-induced convergence. Our work asks a complementary question: given a fixed, imbalanced dataset, can genealogical knowledge guide training strategies to better utilize existing data? Our results suggest yes. Organizing training by language family does not replace computation with human intuition; rather, it reorients the model's attention toward shared phonetic patterns, allowing low-resource languages to benefit from related higher-resource ones. The consistent improvements across LID, CER, and ARI indicate that input structure and not merely data volume, has measurable consequences for what a model learns.

5. Conclusions

This work demonstrates that thoughtful data curation can be a powerful tool for SSL model pretraining, even at modest scales. Although 60 hours of audio is far below the data regimes typically associated with SSL training, our results reveal that structuring the training corpus along linguistic family boundaries yields remarkably capable representations on languages never seen during pretraining. This suggests that the quality and organization of input data can partially compensate for its quantity, a finding with meaningful implications for the preservation and democratization of speech technology across the world's languages. Language remain as the primary medium through which humans share knowledge and ideas, and making robust speech systems accessible beyond high-resource languages remains a critical open challenge.

It is also worth noting that the WavLM Large baseline, without any continued pretraining, already achieved strong performance across downstream tasks, underscoring the generalization capacity of large pretrained models even in zero-shot cross-lingual settings. The fact that family-aware continued pretraining improved over this already strong baseline suggests that the ceiling for this approach has not yet been reached and that larger, more carefully curated corpora could yield substantially stronger representations.

We do not claim these results as universal law. Indeed, recent studies have shown that languages often cluster more cleanly by geographic proximity and areal features than by strict genealogical descent [40]. This suggests that linguistic relatedness is multidimensional, and our family-based groupings may capture only one axis of similarity. Future work will expand this investigation by training using family-based batches composed of different languages and by combining language family structure with geographic proximity to further understand how structural and spatial factors jointly shape learned representation

6. Generative AI Use Disclosure

Large language models (LLMs) were used as coding assistants and to proofread intermediate drafts of this paper. All generated code was reviewed and verified by the authors. The conceptual, methodological, and experimental work was carried out independently by the authors

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A. Data Distribution Details

This appendix provides detailed statistics on the audio data used in our experiments, including total hours, language-wise distribution, and speech duration breakdowns.

Table 3: *Genetic language family mapping. Languages are reported using ISO-639-3 codes.*

Family	**Subfamilies**	**Languages (ISO-639-3)**
Indo-European	Germanic; Romance; Slavic; Baltic; Indo-Iranian; Celtic; Hellenic; Armenian; Albanian	en, de, nl, af, sv, da, nn, no, is, fo, yi; es, fr, it, pt, ro, ca, gl, oc, la; ru, uk, pl, cs, sk, sl, hr, sr, bg, bs, mk, be; lt, lv; ur, hi, bn, mr, pa, ne, si, as, gu, sd, sa, fa, ps; cy, br; el; hy; sq
Afroasiatic	Semitic; Cushitic	ar, he, am, mt; so
Sino-Tibetan	Sinitic; Tibetic; Burmish	zh, yue; bo; my
Dravidian	–	ta, te, kn, ml
Austroasiatic	–	km, vi
Kra–Dai	–	th
Austronesian	–	id, jw, ms, tl, mi, haw
Turkic	–	tr, az, kk
Japonic	–	ja
Koreanic	–	ko
Uralic	–	hu, fi, et
Kartvelian	–	ka
Niger–Congo	–	sw, yo, sn
Mongolic	–	mn
Creole (non-genetic)	–	ht
Language isolate	–	eu

Table 4: *Distribution of validation dataset used during training*

Lang	**Files**	**Speech (min)**	**Speech (hours)**
az	3	155.74	2.60
en	1	68.54	1.14
he	3	101.44	1.69
hu	3	83.79	1.40
ms	3	49.55	0.83
no	2	55.12	0.92
pl	3	102.00	1.70
so	3	139.28	2.32
te	3	93.92	1.57
tl	3	90.50	1.51
Total	27	939.88	20.79

Table 5: *Distribution of "language-aware" dataset used during training*

Lang	**Files**	**Speech (min)**	**Speech (hours)**
am	2	74.31	1.24
ar	2	41.69	0.69
bo	2	220.69	3.68
el	2	86.42	1.44
et	2	23.87	0.40
eu	2	110.53	1.84
fi	2	88.99	1.48
fo	1	12.67	0.21
haw	2	72.69	1.21
ht	2	80.28	1.34
hy	2	45.19	0.75
ja	2	95.96	1.60
ka	1	17.35	0.29
kk	2	68.15	1.14
km	2	80.90	1.35
kn	1	33.26	0.55
ko	2	109.92	1.83
lt	2	43.37	0.72
mi	2	96.71	1.61
ml	2	41.95	0.70
mn	2	134.60	2.24
mt	2	57.09	0.95
my	2	78.30	1.31
pt	2	74.28	1.24
sn	2	31.90	0.53
sw	2	27.66	0.46
ta	2	114.29	1.90
tr	2	117.88	1.96
vi	2	85.97	1.43
yo	2	51.85	0.86
yue	2	122.24	2.04
zh	2	94.81	1.58
Total	63	3355.27	55.92